

KALAJOKI ULKOKALLA, Finland

WMO: 029070

Lat: 64.331N

Lon: 23.446E

Elev: 16

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.5	3.0	-7.2	3.7	-3.8	-0.3	5.4	3.1	39.4	30.2	36.6	31.6	8.8	150	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	70.8	66.5	67.9	63.9	65.6	62.2	66.9	70.4	64.5	67.2	62.5	65.3	10.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	93.5	69.5	63.1	86.6	66.8	61.1	80.6	64.6	31.5	70.8	29.6	67.4	28.2	65.4	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.8	32.4	29.5	-2.7	72.7	9.6	4.3	-9.6	75.7	-15.2	78.2	-20.5	80.6	-27.5	83.7		
(5)			-3.2	67.1	9.4	3.0	-9.9	69.3	-15.4	71.0	-20.7	72.7	-27.6	74.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	24.1	21.9	25.4	32.9	40.9	50.6	59.9	59.5	52.9	43.2	36.2	31.1		
	DBStd	14.38	9.25	10.48	7.81	3.79	4.64	4.58	5.15	4.52	4.17	4.82	5.64	7.26		
	HDD50	4432	802	788	763	513	283	46	1	1	19	214	415	587		
	HDD65	9152	1267	1208	1228	963	746	431	176	177	362	677	865	1052		
	CDD50	779	0	0	0	0	2	65	307	296	107	2	0	0		
	CDD65	24	0	0	0	0	0	0	16	8	0	0	0	0		
	CDH74	18	0	0	0	0	0	0	11	7	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	15.3	17.6	16.1	14.1	12.3	12.9	12.8	11.7	13.9	15.9	17.9	19.3	19.7		
	PrecAvg	29.2	2.6	2.1	1.9	1.6	1.9	2.5	3.1	3.0	2.4	2.6	2.7	2.9		
(15) Precipitation	PrecMax	37.6	5.1	4.0	3.2	3.5	3.7	4.4	5.0	4.8	4.7	4.9	5.4	4.6		
	PrecMin	22.4	0.7	0.2	0.4	0.5	0.5	0.9	0.8	1.1	1.0	0.6	0.3	1.1		
	PrecStd	3.7	1.1	1.0	0.7	0.8	0.9	0.9	1.2	1.1	1.2	1.1	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	37.6	38.4	44.3	55.8	63.9	75.6	74.3	63.2	54.0	47.8	41.6		
		MCWB	36.3	35.4	35.4	40.6	52.8	59.1	69.1	69.1	59.2	51.4	45.8	39.8		
	2%	DB	36.5	35.8	36.5	40.6	51.8	61.1	72.2	69.8	61.2	52.4	46.2	40.5		
		MCWB	35.3	34.0	34.3	37.5	48.7	57.4	68.0	64.9	58.0	50.4	44.8	39.0		
	5%	DB	35.4	34.2	35.4	38.6	49.2	59.0	69.5	67.3	59.8	50.9	44.8	39.6		
		MCWB	34.3	33.0	33.6	36.2	46.6	56.1	65.5	63.3	57.0	49.0	43.4	38.2		
	10%	DB	34.4	33.2	34.3	37.3	46.8	56.9	67.0	65.4	58.4	49.5	43.2	38.5		
		MCWB	33.4	32.2	32.9	35.3	44.4	54.3	63.5	61.9	56.0	47.5	41.7	37.2		
	0.4%	WB	37.2	35.8	36.0	41.3	54.2	60.3	70.4	69.4	60.2	52.3	46.4	40.5		
		MCDB	37.6	37.0	37.2	43.0	54.5	62.3	73.7	73.8	61.9	53.2	47.3	40.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	34.3	34.8	37.8	50.0	58.2	68.3	65.7	58.7	50.8	45.1	39.5		
		MCDB	36.1	35.1	35.9	39.8	51.4	60.3	71.9	68.7	60.5	51.9	46.0	40.2		
	5%	WB	34.5	33.3	33.9	36.6	46.9	56.6	65.7	63.8	57.5	49.3	43.7	38.4		
		MCDB	35.2	34.1	34.9	38.2	48.7	58.7	68.9	66.6	59.3	50.6	44.7	39.3		
	10%	WB	33.5	32.4	33.1	35.6	44.6	54.5	63.8	62.3	56.3	47.8	42.0	37.3		
		MCDB	34.3	33.2	34.1	37.0	46.5	56.5	66.6	65.0	58.2	49.1	43.0	38.3		
	5% DB	MDBR	6.2	6.7	7.4	5.7	5.6	5.3	5.3	4.9	4.3	4.1	4.3	4.8		
		MCDBR	4.6	5.2	5.4	6.8	9.6	7.6	7.9	6.9	5.4	4.4	4.3	4.5		
		MCWBR	4.5	4.9	4.7	5.1	7.0	5.6	5.4	4.8	4.4	4.2	4.2	4.3		
		MCWBR	4.6	5.1	5.0	6.2	9.3	7.0	7.3	6.1	4.9	4.3	4.3	4.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.7	4.8	4.5	4.9	7.5	5.9	5.1	4.9	4.3	4.3	4.4	4.3		
	taub		0.220	0.282	0.308	0.346	0.348	0.351	0.370	0.356	0.332	0.311	0.348	0.350		
		taud	1.995	2.128	2.171	2.217	2.437	2.488	2.457	2.486	2.531	2.369	2.559	2.569		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		105	194	243	261	274	275	265	259	242	188	96	47		
		Edh at Noon	8	19	29	35	31	30	30	27	21	16	7	4		
	RadAvg		37	215	709	1248	1690	1890	1708	1285	715	257	53	14		
(45) All-Sky Solar Radiation	RadStd		4	45	83	110	151	155	151	109	74	46	9	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page